Ashley Taken

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Professional Summary

Machine Learning Engineer and Data Scientist with 6+ years of experience owning ML systems end to end from raw, high frequency data through feature engineering, model development, and evaluation to deployment, monitoring, and ongoing operation. Strong Python, PyTorch, SQL, Docker, and ML engineering background with experience across deep learning, reinforcement learning, time series modeling, and production LLM/RAG systems.

Experience

Data Scientist | Occidental Petroleum (OXY)

Remote | May 2024 – Present

- Led end-to-end development and deployment of a production 2D CNN for well status classification which improved accuracy from 77% to 98% through time series feature engineering, class imbalance handling, confidence thresholds with live inference and ongoing drift monitoring.
- Architected, deployed, and own an enterprise LLM/RAG ecosystem of 22 production AI assistants supporting 8 offshore assets with live-synced knowledge bases, hybrid retrieval, specialized agent orchestration, and custom MCP tools for real time operational data access.
- Own CNN training and inference workflows using focal loss, class weighting and confidence thresholds, with ongoing production data drift monitoring guiding model improvements.
- Engineered and deployed full stack production applications using Python, Flask, Dash, and SQL which integrated with AVEVA PI AF and enterprise IIS infrastructure to automate workflows and save 2 hours per engineer weekly.
- Created and continuously improve 25+ production manage by exception workflows across Seeq, Python, Power Automate, Azure, and AVEVA PI for automated offshore surveillance.

Founder/Machine Learning Engineer | Taken Group LLC

Remote | Jan 2024 – Present

- Developed and operate a full stack quantitative financial research platform using Python, FastAPI, SvelteKit, MariaDB, Redis, and Docker for market data ingestion, feature engineering, model training, backtesting, and experiment management.
- Designed a reproducible experimentation and statistical analysis framework for high frequency time series data incorporating configurable parameter studies, statistical diagnostics, sensitivity analysis, leakage aware validation, and out-of-sample evaluation to support rigorous model and methodology selection.
- Architected a multi-agent AI research and codebase analysis platform using LangGraph, Open WebUI, ChromaDB, hybrid retrieval, cross encoder reranking, and MCP tools for autonomous research and codebase interaction.
- Developed LLM and retrieval evaluation workflows using Recall@K, MRR, groundedness, and LLM-as-judge scoring to measure retrieval accuracy and response quality.
- Implemented parallelized model training and experimentation pipelines using multiprocessing, GPU accelerated PyTorch, and Optuna to execute model comparisons and hyperparameter studies 3x faster.

Senior Process Engineer | Freepoint Eco Systems

Houston, TX | Jan 2024 – May 2024

- Led data architecture planning and implementation for a greenfield pyrolysis facility by designing data flows and management reporting while coordinating vendors and stakeholders to integrate industrial control systems, a data historian, and analytics platforms.
- Facilitated key teamwork and extensive planning workshops between DCS and Historian vendors and Freepoint stakeholders to develop a reliable unit startup plan that prevented up to 5 weeks of potential startup delays.

Implementation Engineer (Forward Deployed Engineer) | Imubit

Houston, TX | Sep 2022 – Jan 2024

- Owned customer facing ML deployments end to end from POC through production, deploying five closed loop reinforcement learning models that delivered $2M+ in customer value.
- Owned post deployment model performance through continuous monitoring, troubleshooting, and optimization of live models using Python, SQL, OSI PI, Seeq, Grafana, and AWS data pipelines.

- Iterated through hundreds of model configurations using statistical and manual evaluation of raw industrial time series data, hyperparameter optimization, and SHAP analysis to select and validate production models.
- Built customer-facing dashboards and model insight tools using Grafana and AWS S3-backed data pipelines to improve model transparency, adoption, and operational trust.

Chevron Phillips Chemical

Baytown, TX | June 2020 – Sep 2022

Advanced Process Controls Engineer – Real-Time Optimization | June 2022 - Sep 2022
- Automated real time furnace monitoring with Python by generating custom status signals and operator alerts to improve response time.
- Developed and owned a production pricing model from development through deployment, improving unit profitability by as much as $300/hr.

Operations Engineer – Olefins | June 2021 – June 2022
- Enhanced distillation tower quality monitoring and production optimization using Python regression models, improving product purity by 0.25 percentage points to strengthen competitive positioning.
- Played a critical role in a 52-day turnaround: conducted tower inspections, led 20+ PSSRs, and delivered post turnaround optimization plans one of which shaved 2 days off of a tower regen bed replacement.

Process Engineer – Polyethylene | June 2020 – June 2021
- Created, productionized, and owned a live linear regression based emissions monitoring and forecasting system used by control room operators across three plants to proactively manage EPA compliance during startups.
- Identified $350K/year in N_2 savings by analyzing a closed-loop regeneration system using condition-based data slicing.

Education

M.S. in Analytics (OMSA - Computational Track) — Georgia Institute of Technology | Dec 2025
Bachelor of Science in Chemical Engineering — University of Houston | May 2020
Bachelor of Science in Biology with Minor in Psychology — Texas A&M University | May 2014

Publications

SPE-226758-MS, *Automated Well Status Identification in US Offshore Operations Using Machine Learning*, Co-Author, SPE Offshore Europe 2025.
- Contribution: Developed XGBoost and 2D CNN models with engineered features to improve offshore well-status classification accuracy**.**

Technical Skills

- **Programming:** Python, SQL, Pandas, NumPy, Scikit-learn, PyTorch, FastAPI, SvelteKit
- **ML/Deep Learning:** XGBoost, CNNs, Transformers, reinforcement learning, time series, SHAP, Optuna
- **ML Engineering:** real time inference, multiprocessing, GPU accelerated training, model monitoring, data drift
- **LLM/RAG:** RAG, LangGraph, ChromaDB, hybrid retrieval, BM25, cross encoder reranking, MCP
- **Platforms/MLOps:** Azure, AWS, Docker, Redis, MariaDB, Grafana, Seeq, AVEVA PI

Leadership & Involvement

- Oxy Production Engineering Community of Practice | *Content Steward* | Aug 2025 - Present
- CPChem PRIDE | Co-Founder & Outreach Coordinator | Jan 2021 – Jul 2022
- CPChem Recruiting Team | College Recruiter | Aug 2021 – Aug 2022